

BRYAN HO

SOFTWARE ENGINEER

Los Angeles, CA | (714) 856-4257 | bryan.ho16@gmail.com | linkedin.com/in/bryan-ho16 | bryannho.com

WORK EXPERIENCE

PLOOMBER | Software Engineer | Remote

September 2023–Present

- Designed and implemented a credits-based billing system for cloud applications, integrating **Stripe** for usage-based payments and enabling flexible resource allocation and pay-as-you-go options for different user tiers
 - Designed a process for uploading pre-set example applications utilizing AWS **S3**, which streamlined customer onboarding, reduced upload time and costs on the backend, and ultimately increased customer retention.
 - Developed bug fixes, improvements to **CI/CD** workflows, and reduced tech-debt for a full-stack distributed web application utilizing **React**, **FastAPI**, and **AWS**
 - Strengthened security protocols by implementing **Auth0** integration and employing encrypted secrets, ensuring robust protection for user web applications.
 - Orchestrated the deployment of over **500** concurrent user web applications using AWS **Elastic Container Service** (ECR), optimizing performance and scalability.
 - Authored and maintained comprehensive documentation for the flagship product and multiple open-source libraries, serving a user base of over **10,000**, enhancing user experience and accessibility.
-

SKILLS

PROGRAMMING LANGUAGES: Python | JavaScript | TypeScript | Java | SQL

FRAMEWORKS: FastAPI | Flask | React | Node.js | SQLAlchemy | Pytest | PyTorch | Scikit-Learn | HTML5 | CSS

TOOLS: AWS | Docker | MySQL | PostgreSQL | NoSQL | MongoDB | Git | Linux

PROJECTS

AUTOMATED ESSAY SCORING AND FEEDBACK GENERATOR | Artificial Intelligence Application | Python
Utilized various natural language processing, machine learning, and deep learning strategies to develop and evaluate several essay evaluation models and provide feedback to English language learners.

- Designed a custom essay evaluation model that combined a **Concurrent Neural Network**, **Long Short-Term Memory Model**, and **Attention Mechanism** to achieve **27%** improved performance over baselines
- Built a feedback model that provided actionable spelling and grammar critiques which increased essay scores by **2.4%**
- Trained and evaluated 4 baseline machine learning and deep learning essay evaluation models including **Neural Networks**, **Regression Models**, and **Attention Models**

VIBESET | Full-stack Application | JavaScript

Full-stack web application that creates Spotify playlists customized by duration, music genre, and energy level at specified points.

- Integrated the **Spotify API** for user authentication, playlist generation, and profile sharing, enhancing user engagement.
 - Developed a full-stack web application using the MERN stack **MongoDB**, **Express**, **React**, **Node.js**
 - Created an algorithm to dynamically sort playlists by fluctuating energy levels at specified time checkpoints
-

EDUCATION

MASTER OF SCIENCE | Computer Science | University of Southern California | GPA 3.6 | December 2022

BACHELOR OF SCIENCE | Computer Science | University of Southern California | GPA 3.7 | May 2022

RELEVANT COURSEWORK | Data Structures, Object Oriented Design, Machine Learning, Deep Learning, Data Science, Operating Systems, Algorithm Analysis, Natural Language Processing, Databases